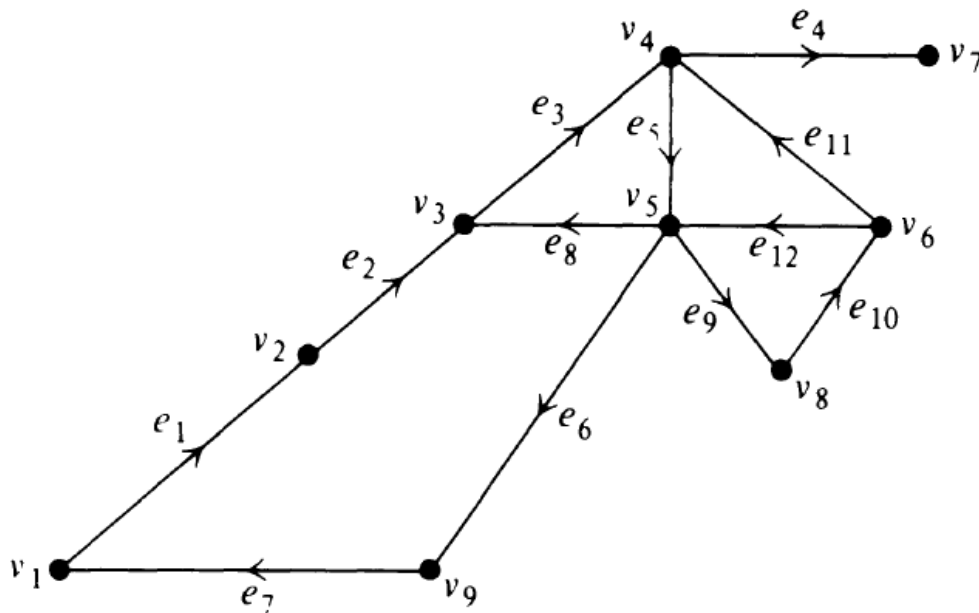


Graph-Euler, Hamiltonian, Planer

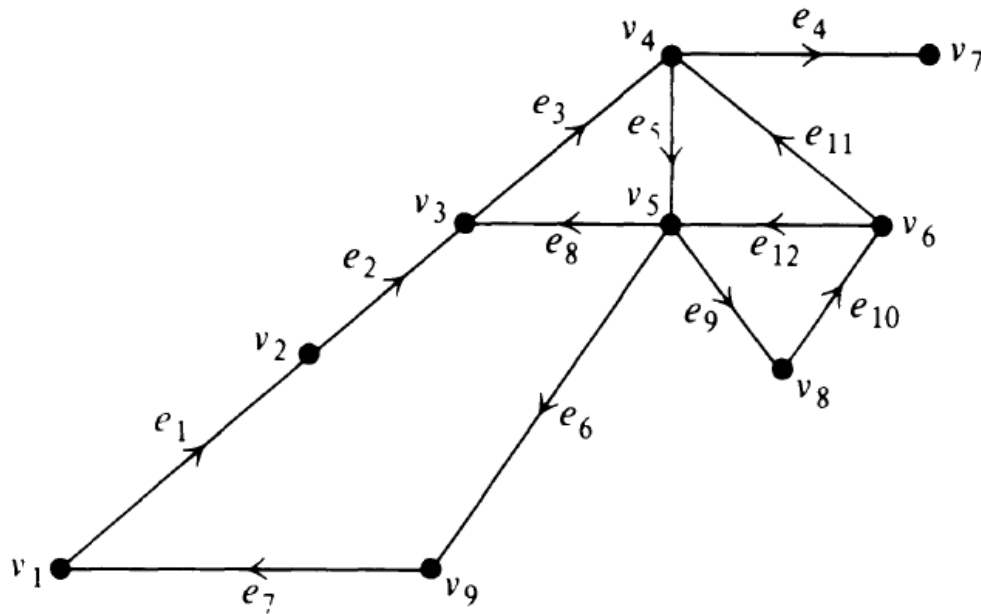
# Path

In a directed graph, a *path* is a sequence of edges  $(e_{i_1}, e_{i_2}, \dots, e_{i_k})^\dagger$  such that the terminal vertex of  $e_{i_j}$  coincides with the initial vertex of  $e_{i_{j+1}}$  for  $1 \leq j \leq k - 1$ . A path is said to be *simple* if it does not include the same edge twice. A path is said to be *elementary* if it does not meet the same vertex twice.



# Circuit

A *circuit* is a path  $(e_{i_1}, e_{i_2}, \dots, e_{i_k})$  in which the terminal vertex of  $e_{i_k}$  coincides with the initial vertex of  $e_{i_1}$ . A circuit is said to be *simple* if it does not include the same edge twice. A circuit is said to be *elementary* if it does not meet the same vertex twice.



**Theorem 5.1** In a (directed or undirected) graph with  $n$  vertices, if there is a path from vertex  $v_1$  to vertex  $v_2$ , then there is a path of no more than  $n - 1$  edges from vertex  $v_1$  to vertex  $v_2$ .

**PROOF** Suppose there is a path from  $v_1$  to  $v_2$ . Let  $(v_1, \dots, v_i, \dots, v_2)$  be the sequence of vertices that the path meets when it is traced from  $v_1$  and  $v_2$ . If there are  $l$  edges in the path, there are  $l + 1$  vertices in the sequence. For  $l$  larger than  $n - 1$ , there must be a vertex  $v_k$  that appears more than once in the sequence, that is,  $(v_1, \dots, v_i, \dots, v_k, \dots, v_k, \dots, v_2)$ . Deleting the edges in the path that leads  $v_k$  back to  $v_k$ , we have a path from  $v_1$  to  $v_2$  that has fewer edges than the original one. This argument can be repeated until we have a path that has  $n - 1$  or fewer edges. □

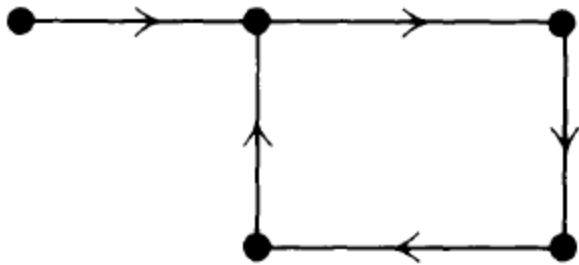
# Connected and Disconnected Graph

An undirected graph is said to be *connected* if there is a path between every two vertices, and is said to be *disconnected* otherwise.

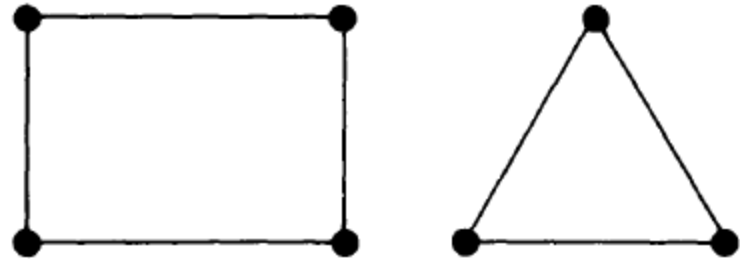
A directed graph is said to be connected if the undirected graph derived from it by ignoring the directions of the edges is connected and is said to be disconnected otherwise.

It follows that a disconnected graph consists of two or more components each of which is a connected graph. A directed graph is said to be *strongly connected* if for every two vertices  $a$  and  $b$  in the graph there is a path from  $a$  to  $b$  as well as a path from  $b$  to  $a$ .

# Connected and Disconnected Graph



(a)



(b)

**a : Connected Graph**

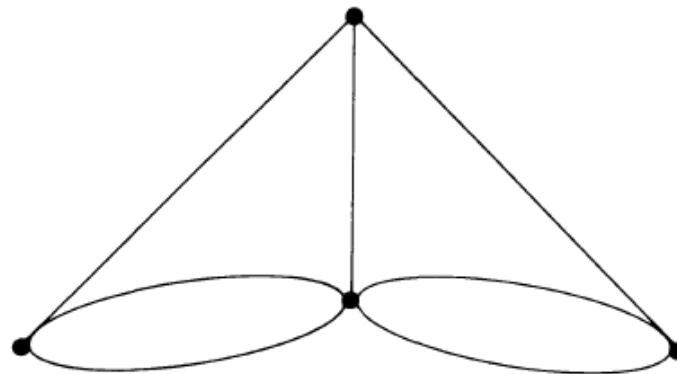
**b : Disconnected Graph**

# Eulerian Paths and Circuits

L. Euler became the father of the theory of graphs when he proved in 1736 that it was not possible to cross each of the seven bridges on the river Pregel in Königsberg, Germany, once and only once in a walking tour. A map of the Königsberg bridges is shown in Fig. 5.14a, and it can be represented as the graph shown in Fig. 5.14b, where the edges represent the bridges and the vertices represent the islands and the two banks of the river.



(a)

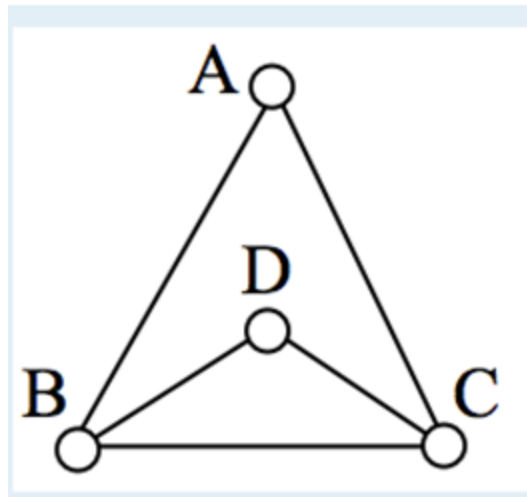


(b)

**Figure 5.14**

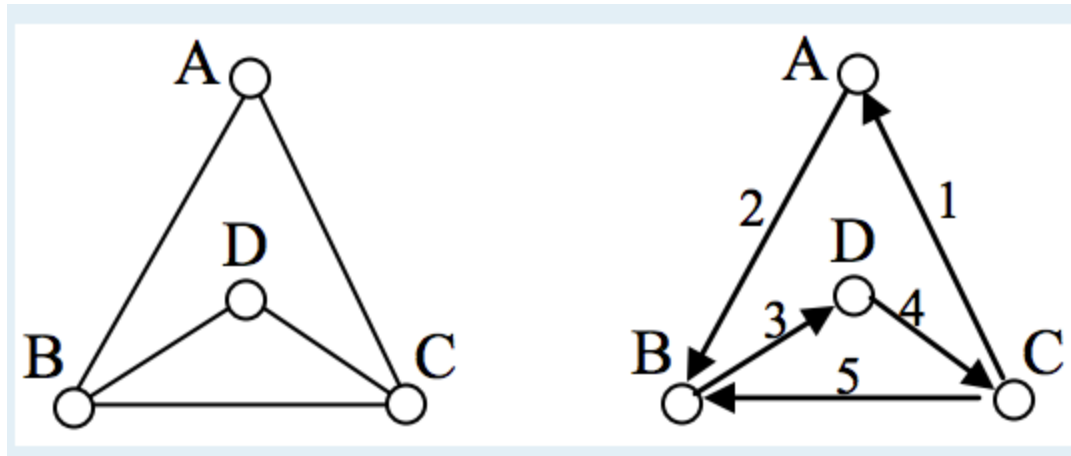
# Euler Path

- An **Euler path** is a path that uses every edge in a graph with no repeats. Being a path, it does not have to return to the starting vertex.



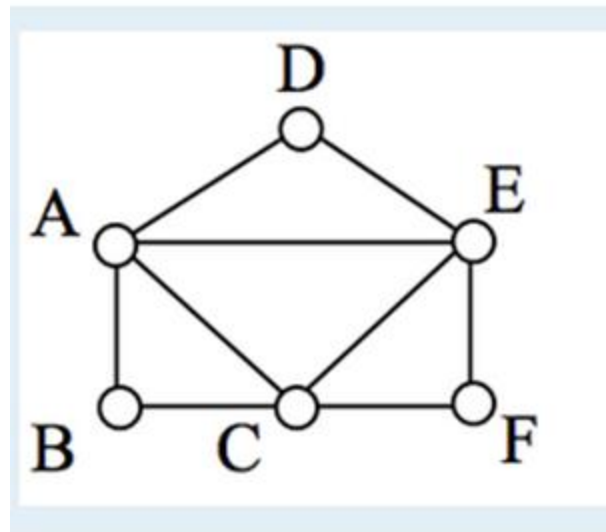


# Euler Path

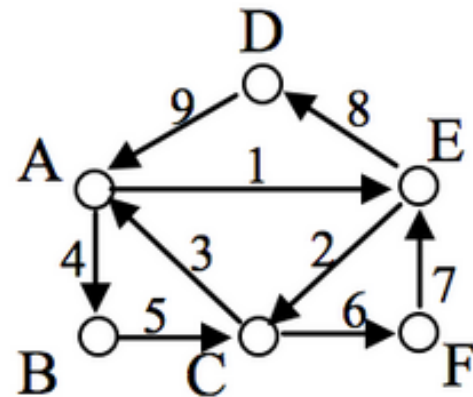
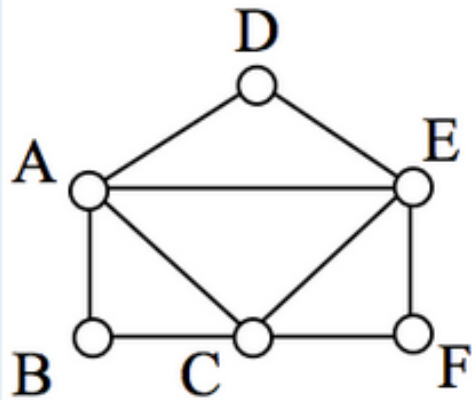


# Euler Circuit

- An **Euler circuit** is a circuit that uses every edge in a graph with no repeats. Being a circuit, it must start and end at the same vertex.



# Euler Circuit



# A result related to Degree of vertices

- In Graph Theory, Handshaking Theorem states in any given graph, Sum of degree of all the vertices is **twice the number of edges contained in it.**

# A result related to Degree of vertices

- Given  $G$  is an undirected graph, the degree of a vertex  $v$ , denoted by  $\deg(v)$ , in graph  $G$  is the number of neighbors of  $v$ .  
The number of vertices of odd degree in any graph  $G$  is even.

# Theorems

- **Euler's Path and Circuit Theorems**
  - A graph will contain an Euler path if it contains at most two vertices of odd degree.
  - A graph will contain an Euler circuit if all vertices have even degree

# **A graph will contain an Euler path if it contains at most two vertices of odd degree.**

- Prove :
  - Graph has zero or two vertices of odd degree(P) is necessary and sufficient condition for it to have eulerian path(Q).
  - $P \leftrightarrow Q$
  - $p \rightarrow q$  : P is sufficient for Q.
  - $q \rightarrow p$  : P is necessary for Q.

**A graph will contain an Euler path if it contains at most two vertices of odd degree.**

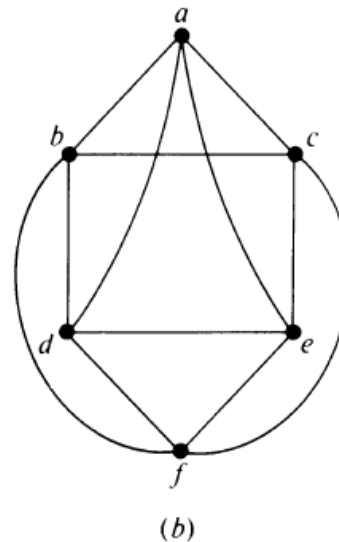
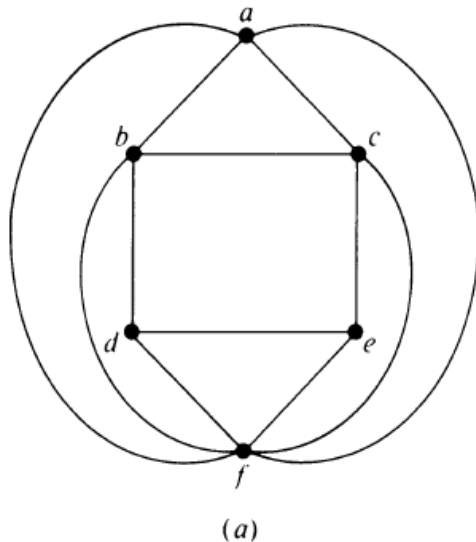
**PROOF** Suppose that the graph possesses an eulerian path. That the graph must be connected is obvious. When the eulerian path is traced, we observe that every time the path meets a vertex, it goes through two edges which are incident with the vertex and have not been traced before. Thus, except for the two vertices at the two ends of the path, the degree of any vertex in the graph must be even. If the two vertices at the two ends of the eulerian path are distinct, they are the only two vertices with odd degree. If they coincide, all vertices have even degree, and the eulerian path becomes an eulerian circuit. Thus, the necessity of the stated condition is proved.



To prove the sufficiency of the stated condition, we construct an eulerian path by starting at one of the two vertices that are of odd degree† and going through the edges of the graph in such a way that no edge will be traced more than once. For a vertex of even degree, whenever the path “enters” the vertex through an edge, it can always “leave” the vertex through another edge that has not been traced before. Therefore, when the construction eventually comes to an end, we must have reached the other vertex of odd degree. If all the edges in the graph were traced this way, clearly, we would have an eulerian path. If not all of the edges in the graph were traced, we shall remove those edges that have been traced and obtain a subgraph formed by the remaining edges. The degrees of the vertices of this subgraph are all even. Moreover, this subgraph must touch the path that we have traced at one or more vertices since the original graph is connected. Starting from one of these vertices, we can again construct a path that passes through the edges. Because the degrees of the vertices are all even, this path must return eventually to the vertex at which it starts. We can combine this path with the path we have constructed to obtain one which starts and ends at the two vertices of odd degree. If necessary, the argument is repeated until we obtain a path that traverses all the edges in the graph. □

# Corollary

- An undirected graph possesses an eulerian circuit if and only if it is connected and its vertices are all of even degree.



**Determine above graphs have euler path or circuit based on degree of vertices.**

# Euler Graph

- Any connected graph is called as an Euler Graph if and only if all its vertices are of even degree.

**OR**

- An Euler Graph is a connected graph that contains an Euler Circuit.

# Directed Graph – indegree and outdegree

- Indegree – Number of edges incident into the vertex.
- Outdegree – Number of edges incident from the vertex.

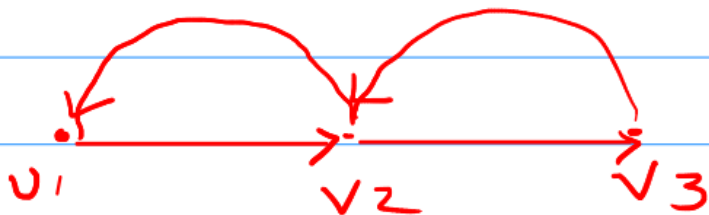
# Theorem

- A directed graph possesses an eulerian circuit if and only if it is connected and the incoming degree of every vertex is equal to its outgoing degree.

# Theorem

- A directed graph possesses an eulerian path if and only if it is connected and the incoming degree of every vertex is equal to its outgoing degree with the possible exception of two vertices. For these two vertices, the incoming degree of one is one larger than its outgoing degree, and the incoming degree of the other is one less than its outgoing degree.

# Example



$$\text{in}(v_1) = 1$$

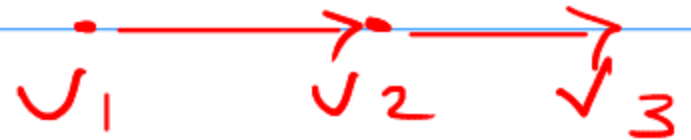
$$\text{out}(v_1) = 1$$

$$\text{in}(v_2) = 2$$

$$\text{out}(v_2) = 2$$

$$\text{in}(v_3) = 1$$

$$\text{out}(v_3) = 1$$



$$\text{in}(v_1) = 0$$

$$\text{out}(v_1) = 1$$

$$\text{in}(v_2) = 1$$

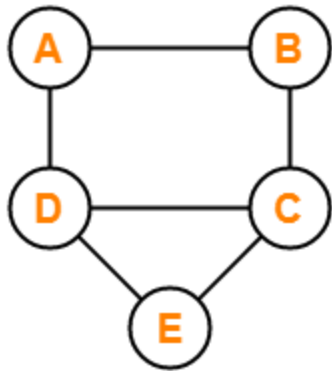
$$\text{out}(v_2) = 1$$

$$\text{in}(v_3) = 1$$

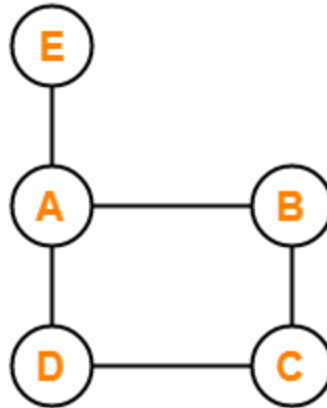
$$\text{out}(v_3) = 0$$

# Hamiltonian Paths and Circuits

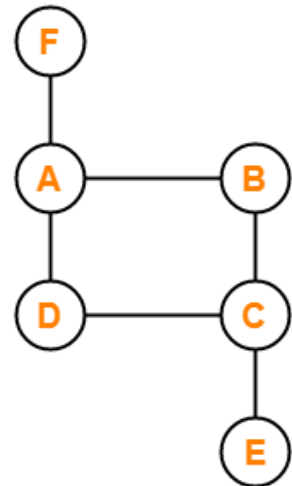
We define a *hamiltonian path* (circuit) to be a path (circuit) that passes through each of the vertices in a graph exactly once.



Hamiltonian Path = ABCDE



Hamiltonian Path = EABCD



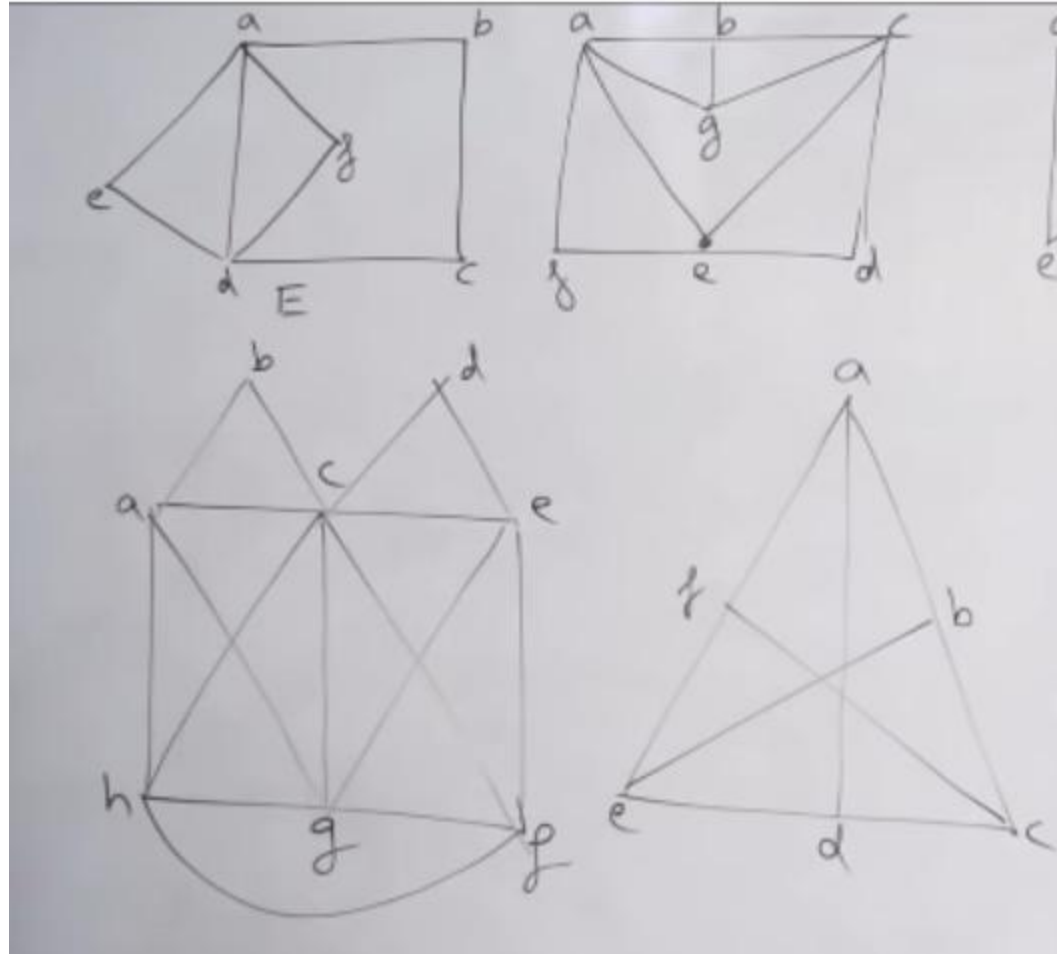
Hamiltonian Path Does Not Exist



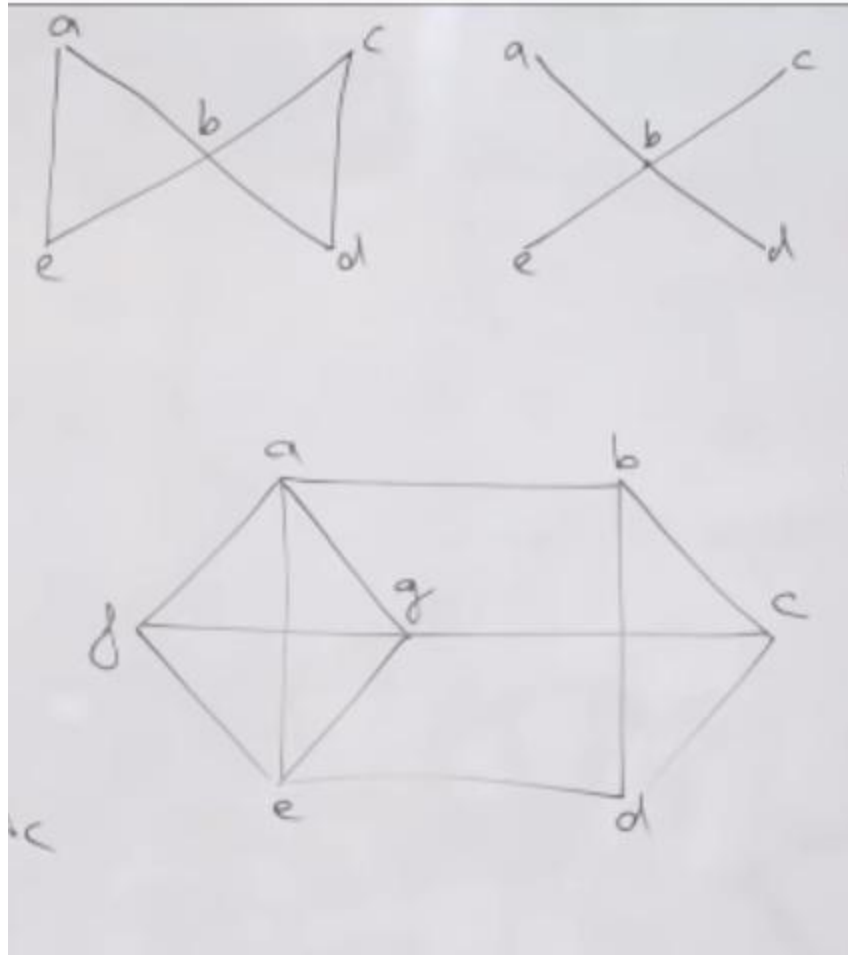
# Hamiltonian Graph

- Any connected graph that contains a Hamiltonian circuit is called as a Hamiltonian Graph.

# Identify : Euler graph and Hamiltonian graph



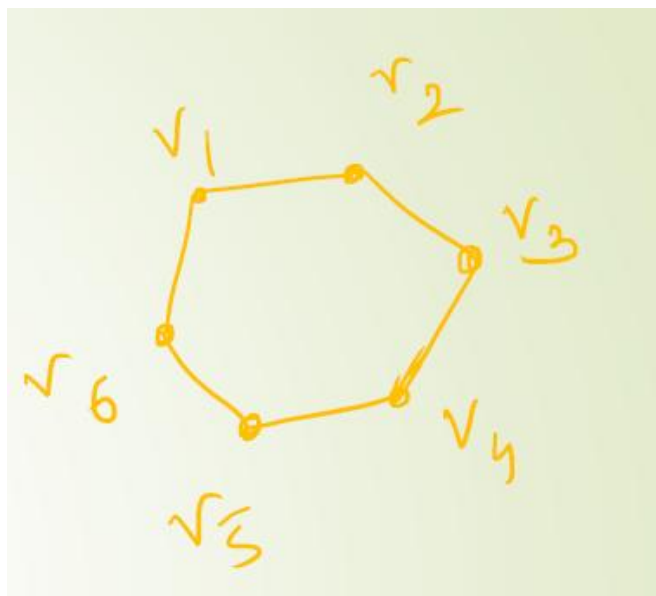
# Identify : Euler graph and Hamiltonian graph



# Theorem

- Let  $G$  be a linear graph of  $n$  vertices. If the sum of the degrees for each pair of vertices in  $G$  is  $n-1$  or larger, then there exists a Hamiltonian path in  $G$ .

It is easy to see that the condition in Theorem 5.4 is a sufficient but not a necessary condition for the existence of a hamiltonian path in a graph. Let  $G$  be an  $n$ -gon,  $n > 5$ . It is clear that  $G$  has a hamiltonian path although the sum of the degrees of any two vertices is 4.

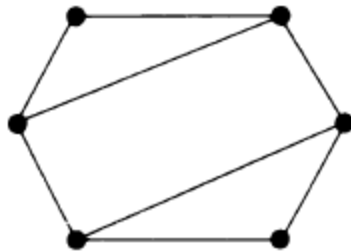


# Theorem

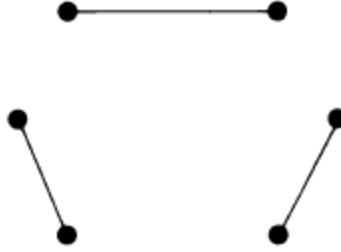
- There is always a Hamiltonian path in a directed complete graph.

# Factors of Graph

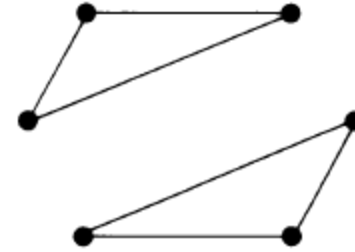
$k$ -factor of a graph is defined to be a spanning subgraph of the graph with the degree of each of its vertices being  $k$ .



(a)



(b)



(c)

a : Original Graph

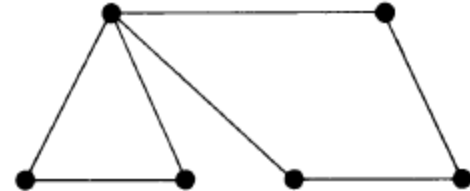
b : 1 Factor Graph

c : 2 Factor Graph

No 1 factor for (a) and no 2 factor for (c) is possible



(a)

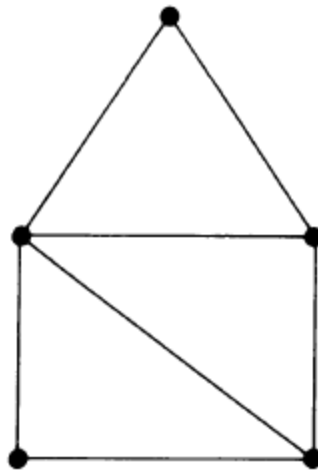


(c)



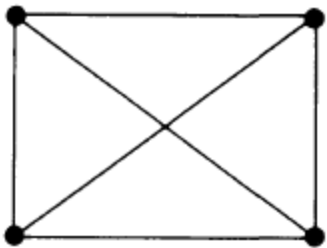
# Planner Graphs

- A graph is said to be planar if it can be drawn in a plane so that no edge cross.

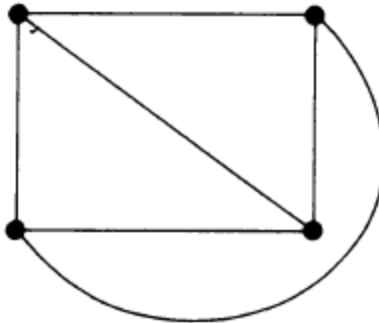


(a)

# Planner Graphs

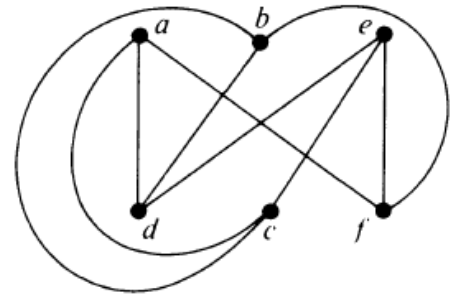


(b)



(c)

(C) is planner version of (b).

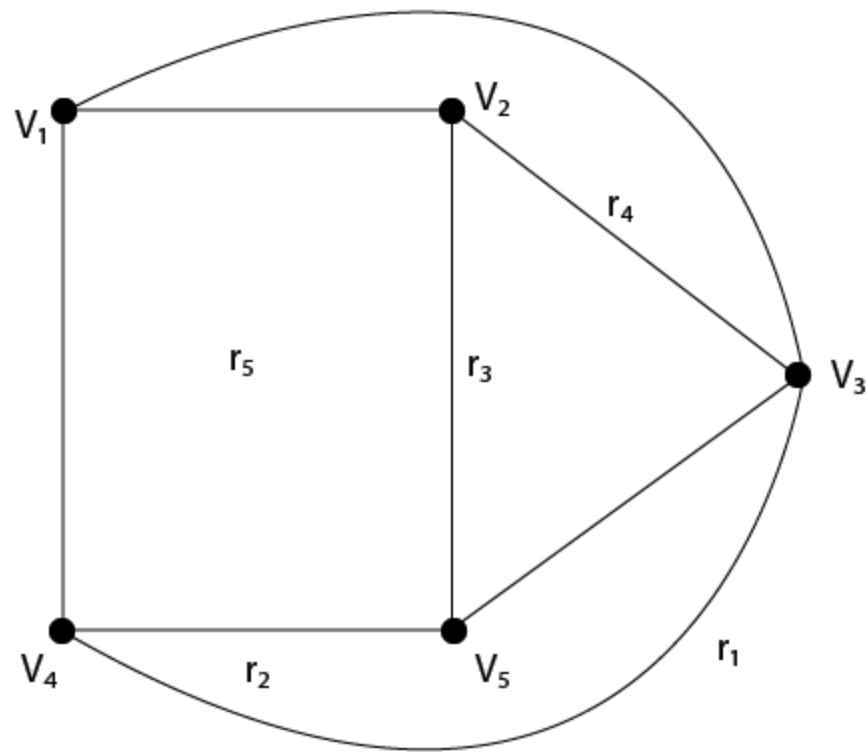


(d)

(d) Is example of non-planner graph

# Region

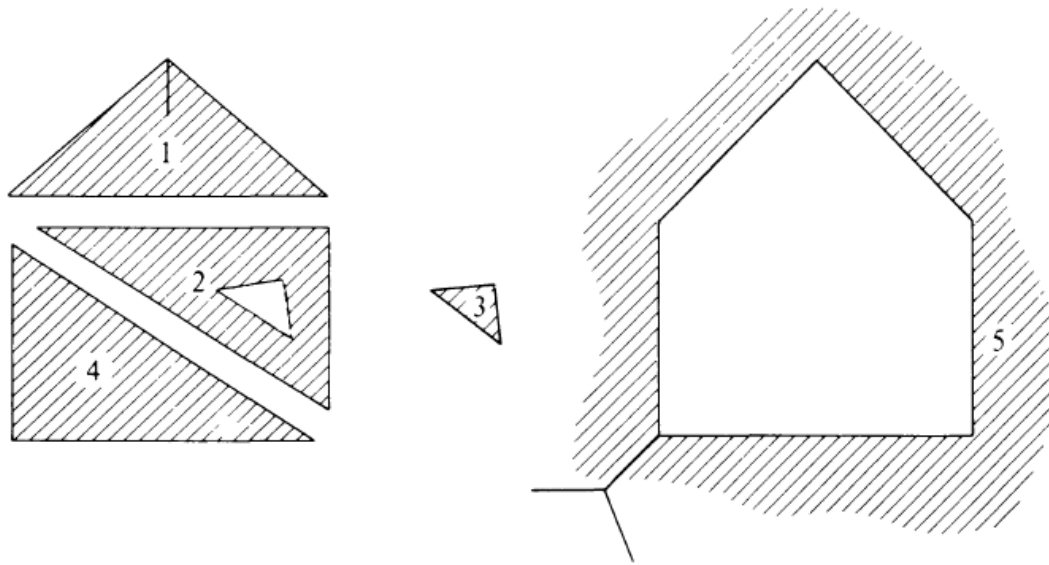
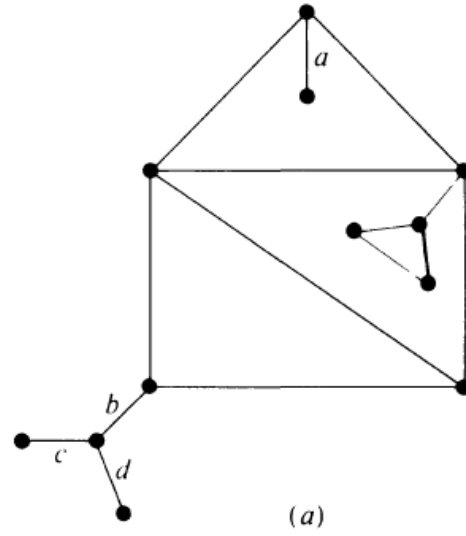
- **Region of a Graph:** Consider a planar graph  $G=(V,E)$ . A region is defined to be an area of the plane that is bounded by edges and cannot be further subdivided. A planar graph divides the plane into one or more regions. One of these regions will be infinite.
- **Finite Region:** If the area of the region is finite, then that region is called a finite region.
- **Infinite Region:** If the area of the region is infinite, that region is called an infinite region. A planar graph has only one infinite region.



**Solution:** There are five regions in the above graph, i.e.  $r_1, r_2, r_3, r_4, r_5$ .

There are four finite regions in the graph, i.e.,  $r_2, r_3, r_4, r_5$ .

There is only one finite region, i.e.,  $r_1$



# Theorem

- If a connected planar graph  $G$  has  $e$  edges,  $v$  vertices, and  $r$  regions, then  $v-e+r=2$ . ( It is also called euler's formula for planar graph)

# Proof

$$v - e + r = 2$$

Proof by Induction

Basis :  $e=1$

$$v_1 \text{ --- } v_2$$

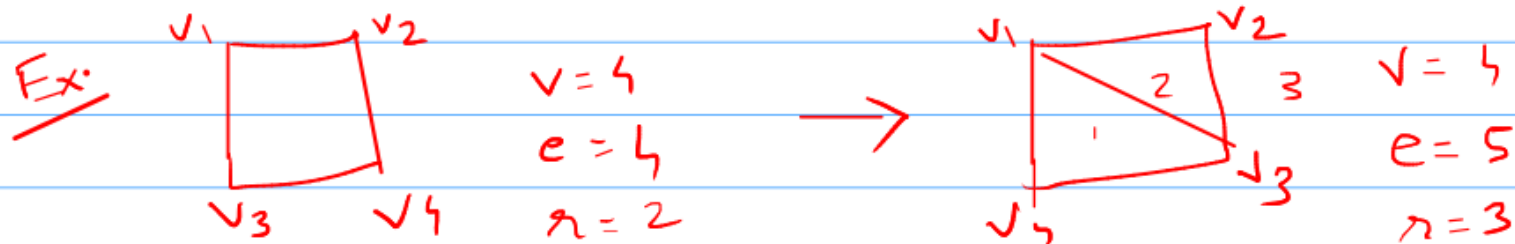
$$v=2, e=1, r=1$$

$$\begin{aligned} v - e + r \\ = 2 - 1 + 1 \\ = 2 \end{aligned}$$

~ Assume for  $e=k$ ,  $v - k + r = 2$  is True.  
└ (i)

~ Let  $e = k+1$

Case-1 New edge is between existing vertices



$$v_k - e_k + r_k = 2$$

Now,  $v_{k+1} = v_k$ ,  $e_{k+1} = e_k + 1$ ,  $r_{k+1} = r_k + 1$

$$v_{k+1} - e_{k+1} + r_{k+1}$$

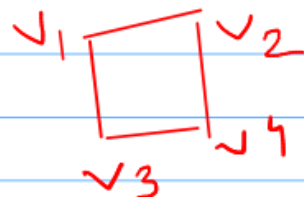
$$\Rightarrow v_k - (e_k + 1) + r_k + 1$$

$$\Rightarrow v_k - e_k + r_k = 2$$

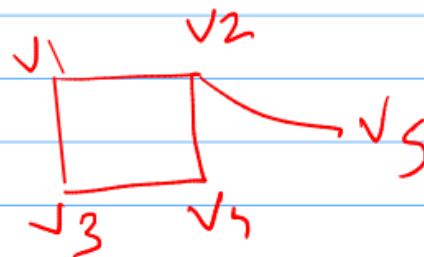
$$\Rightarrow \text{So, } v_{k+1} - e_{k+1} + r_{k+1} = 2$$



Case II



$$V=4 \quad n=2 \\ e=4$$



$$V=5 \quad n=2 \\ e=5$$

$$V_{k+1} = V_k + 1, \quad e_{k+1} = e_k + 1, \quad g_{k+1} = g_k$$

$$\begin{aligned} & V_{k+1} - e_{k+1} + g_{k+1} \\ &= V_{k+1} - (e_k + 1) + g_k \\ &= V_k - e_k + g_k = 2 \end{aligned}$$

$$\text{So, } V_{k+1} - e_{k+1} + g_{k+1} = 2$$

- If a connected planar graph  $G$  with no loops has  $e$  edges and  $v$  vertices ( $v \geq 3$ ), then  $3v - e \geq 6$ .

If a connected planar graph  $G$  with no loops has  $e$  edges and  $v$  vertices ( $v \geq 3$ ), then  $3v - e \geq 6$ .

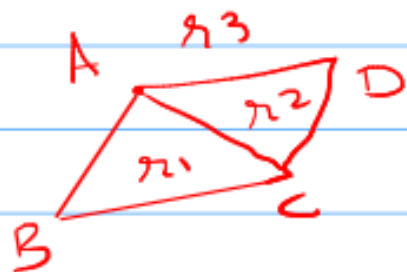
Let  $v = 3$



$$h = 2$$

$$e = 3$$

$$3h = 2e$$



$$h = 3$$

$$3h = 9$$

$$e = 5$$

$$2e = 10$$

$$3h \leq 2e$$

— for all Cases



Now,  $d - e + r = 2$

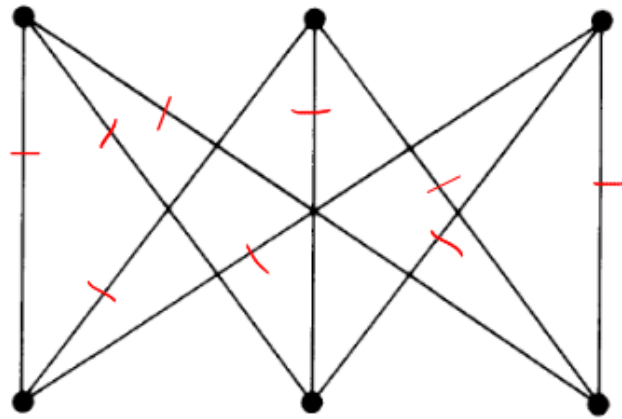
$$d = e - r + 2$$

from (i),  $r \leq \frac{2e}{3}$

So,  $d \geq e - \frac{2e}{3} + 2$

$$d \geq \frac{e+6}{3}$$

$$\boxed{3d - 6 \geq e}$$



$$e = 9$$

$$v = 6$$

$$3v - 6$$

$$= 12$$

$3v - 6 \geq e$  is Satisfied

So, Condition is made more strict

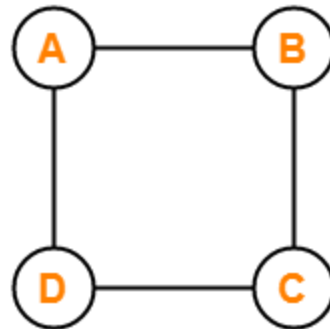
$$2v - 4 \geq e$$

# Bipartite Graph

- A Bipartite Graph is a graph whose vertices can be divided into two independent sets,  $U$  and  $V$  such that every edge  $(u, v)$  either connects a vertex from  $U$  to  $V$  or a vertex from  $V$  to  $U$ . In other words, for every edge  $(u, v)$ , either  $u$  belongs to  $U$  and  $v$  to  $V$ , or  $u$  belongs to  $V$  and  $v$  to  $U$ . We can also say that there is no edge that connects vertices of same set.

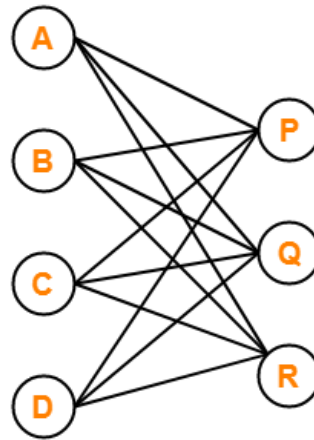
# Bipartite Graph

- The vertices of the graph can be decomposed into two sets.
- The two sets are  $X = \{A, C\}$  and  $Y = \{B, D\}$ .
- The vertices of set  $X$  join only with the vertices of set  $Y$  and vice-versa.
- The vertices within the same set do not join.
- Therefore, it is a bipartite graph.



# Complete Bipartite Graph

- A bipartite graph where every vertex of set  $X$  is joined to every vertex of set  $Y$  is called as complete bipartite graph.

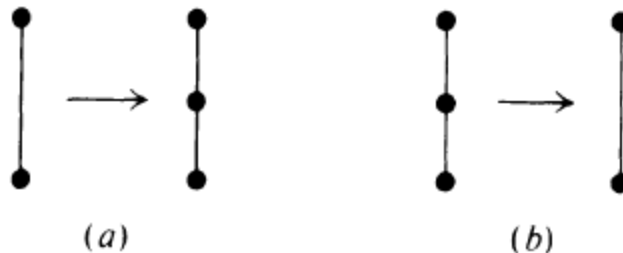


Example of Complete Bipartite Graph

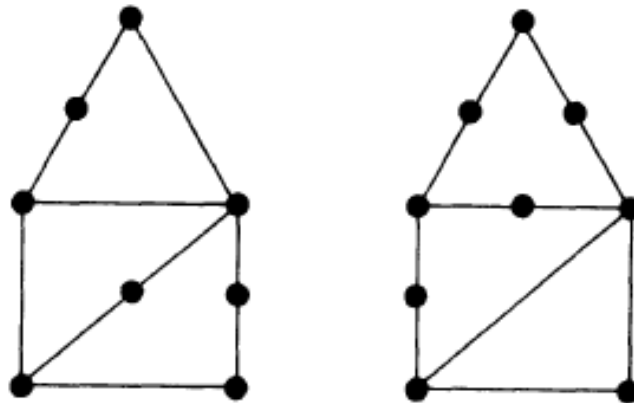


# Isomorphic to within vertices of degree 2

- Two graphs  $G_1$  and  $G_2$  are said to be isomorphic to within vertices of degree 2 if they are isomorphic or if they can be transformed into isomorphic graphs by repeated insertion and/or removal of vertices of degree 2.



# Isomorphic to within vertices of degree 2

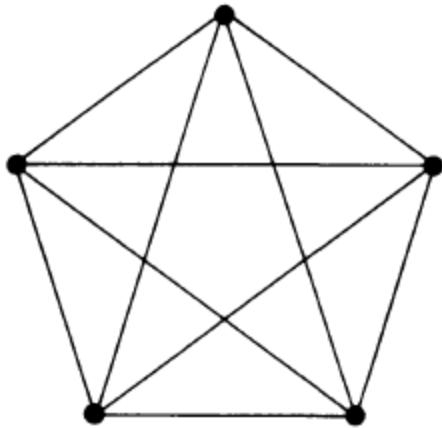


(c)

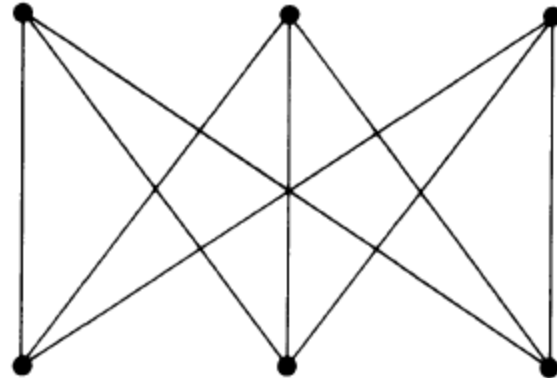
# ***Kuratowski Graph***

- Below two graphs are called Kuratowski graphs.
  - $K_5$  (the complete graph on five vertices)
  - $K_{3,3}$  (complete bipartite graph on six vertices, three of which connect to each of the other three)

# ***Kuratowski Graph***



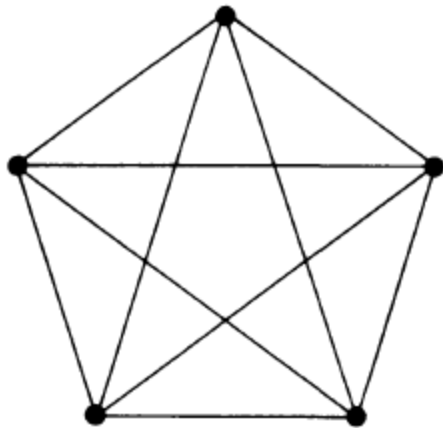
**Figure 5.34**



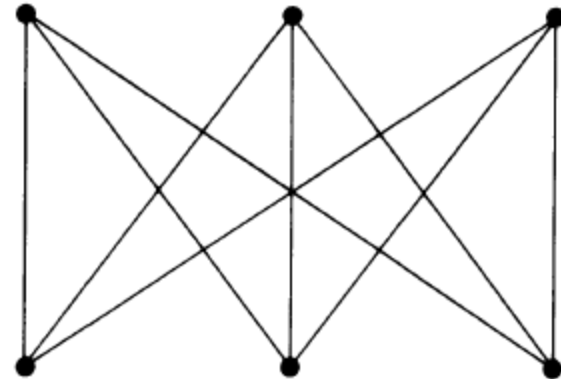
**Figure 5.35**

# ***Theorem(Kuratowski)***

**Theorem 5.8 (Kuratowski)** A graph is planar if and only if it does not contain any subgraph that is isomorphic to within vertices of degree 2 to either the graph in Fig. 5.34 or Fig. 5.35.



**Figure 5.34**



**Figure 5.35**